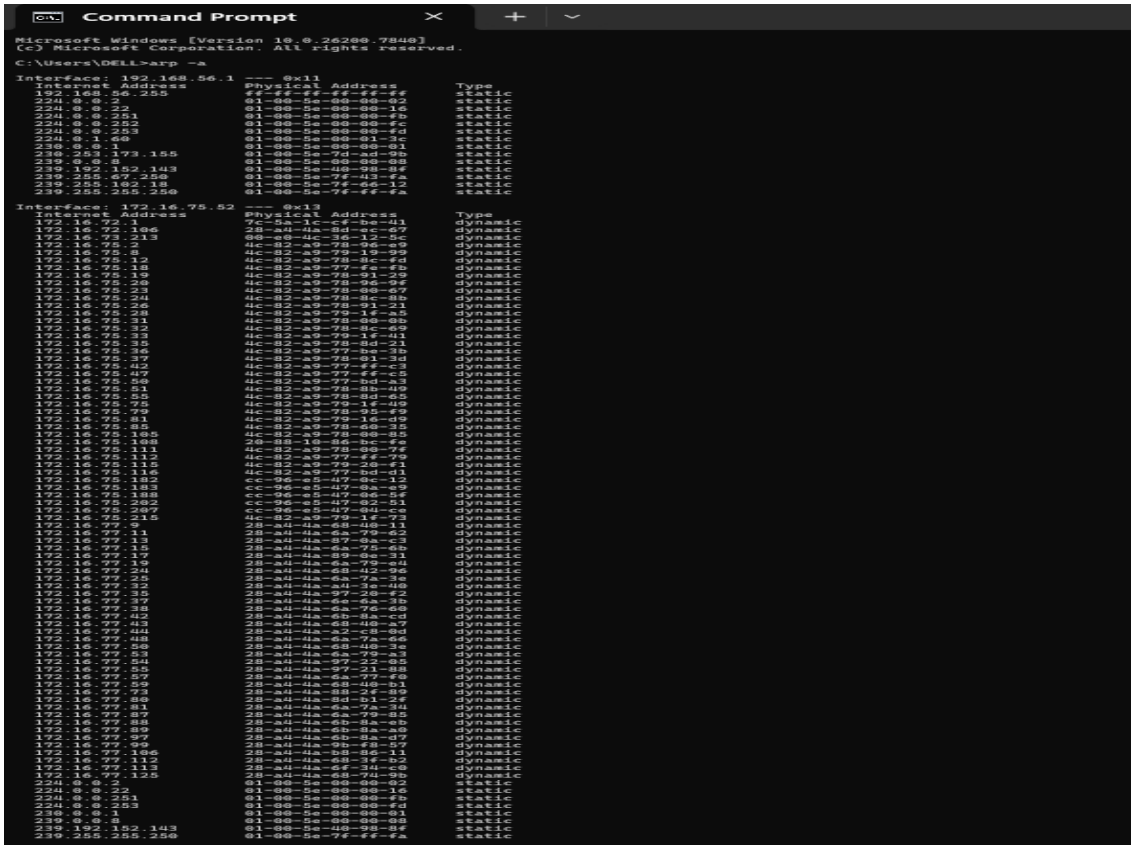
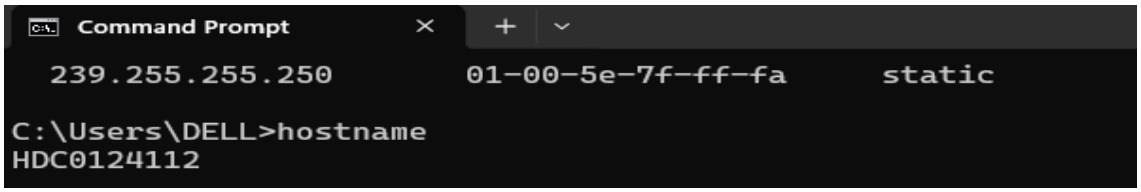





```
C:\Users\DELL>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc42:768d%17
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.75.52
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 172.16.72.1
```

```
C:\Users\DELL>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HDC0124112
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Ethernet Connection (17) I219-LM
   Physical Address. . . . . . . . . : 20-88-10-86-C7-DB
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-11
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc42:768d%17(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 873070631
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-4A-B6-03-20-88-10-86-C7-DB
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 4E-82-A9-78-90-37
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 42-82-A9-78-90-37
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 4C-82-A9-78-90-37
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 172.16.75.52(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 172.16.72.1
   DNS Servers . . . . . . . . . . . : 172.16.72.1
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

```
C:\Users\DELL>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

```
C:\Users\DELL>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

```
C:\Users\DELL>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.

  command      One of these:
                 PRINT     Prints  a route
                 ADD       Adds    a route
                 DELETE    Deletes a route
                 CHANGE    Modifies an existing route
  destination  Specifies the host.
  MASK         Specifies that the next parameter is the 'netmask' value.
  netmask      Specifies a subnet mask value for this route entry.
               If not specified, it defaults to 255.255.255.255.
  gateway      Specifies gateway.
  interface    the interface number for the specified route.
  METRIC       specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.
```

```
C:\Users\DELL>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        HDC0124113:49676       ESTABLISHED
  TCP    127.0.0.1:49676        HDC0124113:49675       ESTABLISHED
  TCP    127.0.0.1:49677        HDC0124113:49678       ESTABLISHED
  TCP    127.0.0.1:49678        HDC0124113:49677       ESTABLISHED
  TCP    127.0.0.1:49682        HDC0124113:49683       ESTABLISHED
  TCP    127.0.0.1:49683        HDC0124113:49682       ESTABLISHED
  TCP    127.0.0.1:63945        HDC0124113:63944       TIME_WAIT
  TCP    172.16.75.53:1521      HDC0124113:57609       ESTABLISHED
  TCP    172.16.75.53:7680      172.16.75.35:49193     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.75.85:60908     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.75.105:58668    ESTABLISHED
  TCP    172.16.75.53:7680      172.16.77.24:55750     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.33:56991     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.34:49585     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.37:59924     TIME_WAIT
  TCP    172.16.75.53:7680      THALAPATHY-VIJA:61829  TIME_WAIT
  TCP    172.16.75.53:49414     4.213.25.242:https     ESTABLISHED
  TCP    172.16.75.53:49964     pnmaaa-bf-in-f10:https  ESTABLISHED
  TCP    172.16.75.53:50044     sa-in-f188:5228        ESTABLISHED
  TCP    172.16.75.53:56658     172.16.75.115:ms-do    ESTABLISHED
  TCP    172.16.75.53:57609     HDC0124113:1521        ESTABLISHED
  TCP    172.16.75.53:59971     172.16.75.115:ms-do    ESTABLISHED
  TCP    172.16.75.53:60774     4.150.223.97:https     ESTABLISHED
  TCP    172.16.75.53:60779     172.16.75.115:ms-do    ESTABLISHED
  TCP    172.16.75.53:60781     172.16.75.116:ms-do    ESTABLISHED
  TCP    172.16.75.53:63934     lcmaaa-an-in-f3:https  TIME_WAIT
  TCP    172.16.75.53:63947     150.171.110.83:https   ESTABLISHED
  TCP    172.16.75.53:63949     172.16.75.105:ms-do    ESTABLISHED
  TCP    172.16.75.53:63951     a23-211-105-64:https   ESTABLISHED
  TCP    172.16.75.53:63952     a23-211-105-64:https   ESTABLISHED
  TCP    172.16.75.53:63958     52.123.128.14:https    ESTABLISHED
  TCP    172.16.75.53:63960     20.42.73.30:https      ESTABLISHED
  TCP    172.16.75.53:63961     20.42.73.30:https      ESTABLISHED
  TCP    172.16.75.53:63962     52.123.129.14:https    ESTABLISHED
  TCP    172.16.75.53:63963     172.16.75.116:ms-do    TIME_WAIT
  TCP    172.16.75.53:63965     172.16.75.99:ms-do     SYN_SENT
  TCP    172.16.75.53:64969     dns:https              CLOSE_WAIT
  TCP    172.16.75.53:65244     172.16.75.216:ms-do    ESTABLISHED
  TCP    172.16.75.53:65246     172.16.77.70:ms-do     ESTABLISHED
```